

Chemotherapy and pregnancy

Most people who have had chemotherapy when they are pregnant have healthy babies. You can usually start chemotherapy after you are 14 weeks pregnant.

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About chemotherapy during pregnancy

[Chemotherapy](#) is the most common [treatment used during pregnancy.](#) The drugs destroy cancer cells, but they also affect healthy cells. It is natural to feel anxious about the possible effects of chemotherapy on the baby. But, at the same time, you may want to start treatment for the cancer.

If your doctors think you need chemotherapy, they may:

- delay chemotherapy until after the [first trimester](#)
- avoid using a certain drug because it is harmful to the baby
- avoid using a certain drug because there is not enough evidence to show it is safe in pregnancy.

Effects of chemotherapy on the baby

Studies have looked at babies whose mothers had [chemotherapy](#) after the first trimester. These studies are generally reassuring. Most women had healthy babies. So far, studies do not show any differences in the baby's development. But doctors are still studying the long-term effects to find out more about any other possible risks.

Risk of earlier delivery

There is some evidence to suggest chemotherapy may increase the risk of giving birth earlier and the baby having a lower birth weight. But doctors now have more experience giving chemotherapy during pregnancy. This means babies are less likely to be born early. If possible, your doctor will try to help your pregnancy go to full term (37 weeks).

Some women have a higher risk of early delivery. In this situation, doctors recommend they are cared for in hospitals with specialist baby units. These are called obstetric high dependency units (OHDUs).

When chemotherapy is given during pregnancy

You do not usually have [chemotherapy](#) during the first trimester. This is because the baby's organs are still forming, and chemotherapy can increase the risk of a miscarriage or birth defect.

You can usually start chemotherapy after you are 14 weeks pregnant (in the second trimester). At this stage, research shows most chemotherapy drugs will not harm the baby. The placenta acts as a barrier between you and the baby. Some drugs cannot pass through the placenta. Others only pass through in very small amounts. Your cancer doctor and specialist nurse will explain this to you.

Having chemotherapy

You will usually have chemotherapy as an injection or a drip (infusion) into a vein, or as tablets. This is the same for someone who is not pregnant.

It may be helpful to talk to someone who has also had chemotherapy during pregnancy. [Mummy's Star](#) may be able to arrange this for you. Your cancer team may also help with this.

When you stop chemotherapy

Your doctors will talk to you about when you will stop chemotherapy. Chemotherapy is not usually given after you are 37 weeks pregnant. You will have a break between your last dose of chemotherapy and your expected delivery date. This is to avoid the baby being born when your blood cell levels are still low. Having a low level of blood cells is a temporary [side effect of chemotherapy](#). It can increase your risk of serious infection and bleeding.

If your baby is born soon after your chemotherapy finishes, doctors can give you drugs to support your immune system. This helps you fight [infections](#).

Different cancers and chemotherapy

Chemotherapy can be given to treat different cancers during pregnancy. These are some examples. Your specialist doctor or nurse will give you information about your individual situation.

Breast cancer and chemotherapy in pregnancy

You may have chemotherapy before or after surgery to remove [breast cancer](#). Doctors will use the same chemotherapy drugs they would give to someone with breast cancer who is not pregnant.

Your doctor will usually give you a combination of:

- [doxorubicin](#) (Adriamycin[®])
- [cyclophosphamide](#)
- [epirubicin](#).

These drugs used to treat breast cancer during pregnancy.

You might have a drug called [docetaxel](#) (Taxotere[®]) or [paclitaxol](#) (Taxol[®]). These are called taxanes. These drugs are less commonly used in pregnancy. You may have them nearer the end of your chemotherapy treatment. This often means you have them after the baby is born.

You might have other treatments after the baby is born. These include [radiotherapy](#), [hormonal](#) and [targeted therapy](#).

Cancer of the cervix and chemotherapy in pregnancy

Doctors will use the same chemotherapy drugs they would give to someone with [cervical cancer](#) who is not pregnant.

They may use the drugs [cisplatin](#) or [carboplatin](#) along with other chemotherapy drugs.

In some situations, chemotherapy may be given so that [surgery](#) can be delayed until the baby has grown enough to be born. You may have further treatment, such as [surgery](#) or [radiotherapy](#), after the baby is born. You may also have more [chemotherapy](#) after the baby is born.

Non-Hodgkin lymphoma (NHL) and chemotherapy in pregnancy

Chemotherapy can be used to treat fast-growing [non-Hodgkin lymphoma \(NHL\)](#) during pregnancy. [CHOP](#) is the standard combination of chemotherapy drugs used. It can be given during pregnancy. It is made up of:

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- C – cyclophosphamide
- H – doxorubicin (hydroxydaunomycin)
- O – vincristine (Oncovin[®])
- P – prednisolone (a steroid).

[Rituximab](#) is a targeted therapy drug usually given with CHOP. This treatment is called [R-CHOP](#). Doctors usually give it to you after the baby is born.

Hodgkin lymphoma (HL) and chemotherapy in pregnancy

[Hodgkin lymphoma \(HL\)](#) is commonly treated using a combination of drugs called [ABVD](#). It can be given during pregnancy. It is made up of:

- A – [doxorubicin](#) (Adriamycin[®])
- B – bleomycin
- V – vinblastine (Velbe[®])
- D – dacarbazine (DTIC).

Acute leukaemia and chemotherapy in pregnancy

Chemotherapy is used to destroy all the cancer cells for [acute myeloid leukaemia \(AML\)](#). This is called induction chemotherapy. The drugs that are usually given in this treatment are [daunorubicin](#) and [cytarabine](#). You can still have these drugs if you are pregnant.

Acute promyelocytic leukaemia (APL)

A rare type of AML called [acute promyelocytic leukaemia \(APL\)](#) is treated in a different way. The first treatment is a drug called ATRA, which is also called tretinoin or Vesanoind[®]. ATRA can reduce leukaemia symptoms very quickly. It is usually given with chemotherapy or a drug called arsenic trioxide. You cannot have ATRA during the first 14 weeks of pregnancy.

APL needs to be treated very quickly. There is a risk of having a serious bleed without treatment. Unless it is diagnosed later in pregnancy, doctors may recommend [ending the pregnancy](#). This is so you can have the most effective treatment. If you decide to continue with the pregnancy, ATRA is usually given on its own.

Getting support

Macmillan is here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
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[Chat to our specialists online.](#)

- Visit our [chemotherapy forum](#) to talk with people who have had chemotherapy, share your experience, and ask questions.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our pregnancy and cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help